$$p\left(\frac{2}{3}\right) = \frac{(120-120)}{9}$$

$$= 0$$

Therefore, $(3x-2)$ is a factor of $3x^3 + x^2 - 20x + 12$

Example 3.15 Find the value of m, if $(x-2)$ is a factor of the polynomial $2x^3 - 6x^2 + mx + 4$.

Solution

Let $p(x) = 2x^3 - 6x^2 + mx + 4$

By factor theorem, $(x-2)$ is a factor of $p(x)$, if $p(2) = 0$

$$\begin{aligned} p(2) &= 0 \\ 2(2)^3 - 6(2)^2 + m(2) + 4 &= 0 \\ 2(8) - 6(4) + 2m + 4 &= 0 \\ -4 + 2m &= 0 \\ m &= 2 \end{aligned}$$

To find the zero of x–2;
put $x - 2 = 0$
we get $x = 2$

Exercise 3.3

1. Check whether $p(x)$ is a multiple of g(x) or not .
   $p(x) = x^3 - 5x^2 + 4x - 3$; $g(x) = x - 2$
2. By remainder theorem, find the remainder when, $p(x)$ is divided by $g(x)$ where,
   (i) $p(x) = x^3 - 2x^2 - 4x - 1$; $g(x) = x + 1$
   (ii) $p(x) = 4x^3 - 12x^2 + 14x - 3$; $g(x) = 2x - 1$
   (iii) $p(x) = x^3 - 3x^2 + 4x + 50$; $g(x) = x - 3$
3. Find the remainder when $3x^3 - 4x^2 + 7x - 5$ is divided by $(x+3)$
4. What is the remainder when $x^{2018} + 2018$ is divided by $x-1$
5. For what value of k is the polynomial $p(x) = 2x^3 - kx^2 + 3x + 10$ exactly divisible by $(x-2)$
6. If two polynomials $2x^3 + ax^2 + 4x - 12$ and $x^3 + x^2 - 2x + a$ leave the same remainder when divided by $(x - 3)$, find the value of a and also find the remainder.
7. Determine whether $(x-1)$ is a factor of the following polynomials:
   i) $x^3 + 5x^2 - 10x + 4$ *ii)* $x^4 + 5x^2 - 5x + 1$

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